RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch	**:** IV Semester B.E CSE
Course Code & Title	**:** CS8492 & Database Management Systems
Name of the Faculty member	**:** Mr. K.Vignesh Saravanan, AP / CSE

Innovative Practice Description

- **Unit / Topic: Unit V / Advanced Topics /** XML Databases: XML Hierarchical Model, DTD, XML Schema, XQuery
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO 13
- **Activity Chosen: Think-Pair-Share**
- **Justification:**
  - Active learning strategy, in which students work on a problem posed by instructor, first individually (Think), then in pairs (Pair) or groups, and finally together with the entire class (Share).
  - **T (Think):** Teacher asks a specific question about the topic. Students "think" about what they know or have learned, and come up with their own individual answer to the question. [Takes 1-3 Minutes].
  - **P (Pair):** Teacher asks another question, related to the previous one that is suitable to deepen the students' understanding of the topic. Each student is paired with another student. They share their thinking with each other and proceed with the task. [Takes 5-10 Minutes].
  - **S (Share):** Students share their thinking (or solution) with the entire class. Teacher moderates the discussion and highlights important points. [Takes 10-20 minutes].
- **Time Allotted for the Activity:** 40 Minutes
- **Details of the Implementation:**
  - Think-Pair-Share innovative practice conducted for II year CSE students, after explanation of the concept of XML Databases: XML Hierarchical Model, DTD, XML Schema and XQuery.
  - First, I asked the students to think about the real time implementation of the XML structures in internet webpages. The students searched few webpages and identified the XML DTD and Schema structures.
  - Then I make them as a pair and share their thinking with their partners.

- Finally I asked any two or three pairs to explain the concept to whole class for further discussion. Mr.Dhiraj and Mr.Sivakumar, II year CSE shared their ideas and knowledge of XML DTD, schema and how to write X-Query and X-Path to all students.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PO12	PSO1
CO1	3	3	3	1	1	1	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PO12
	2	2	3	3	3	2
Justification for correlation	Apply basic Knowledge and fundamentals in XML based databases	Identify the need of XML DTD, Schema, X-query	Able to design and develop XML structure and X-query and X-Path	Functional individually in identifying when to XML databases DTD, Schema	Communicate/ share the ideas with other students in written and oral presentation	Ability to reproduce the contents gathered through self-learning

- **Images / Screenshot of the practice:**


- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - Working as a pair, they can share their knowledge with them and come up with better understanding about the topic.

- Different types of reasoning/justification based questions are discussed in the class room, so students come to know how to answer/draw diagrams whenever new real time system is asked in the question.

❖ ***Benefit of the practice:***

- Students are actively engaged and students learn from each other.
- Makes class interactive and builds a friendly, yet academic atmosphere.
- Includes all the students in the teaching-learning process.

❖ ***Challenges faced in implementation:***

- Students felt very difficult in this activity and could not complete in time whenever new real time system asked in the question.
- Students are hesitating to speak in front of all.

References:

❖ http://www.et.iitb.ac.in/TeachingResources.html

❖ https://www.readingrockets.org/strategies/think-pair-share

❖ https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html

❖ https://serc.carleton.edu/introgeo/interactive/tpshare.html

Signature of Faculty Member **HOD**